

CAIRO INTL, Egypt

WMO: 623660

Lat: 30.122N

Lon: 31.406E

Elev: 116

StdP: 99.93

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	9.2	-4.1	2.7	20.6	-1.9	3.3	20.4	12.0	16.0	10.6	16.1	2.6	90	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.3	38.8	21.0	37.1	21.4	36.0	21.6	25.4	31.6	24.7	31.1	24.2	30.5	5.2	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.0	28.0	23.1	18.1	27.6	22.3	17.3	27.2	78.4	31.8	75.8	31.0	73.7	30.7	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	8.2	7.3	DB	6.1	42.3	1.8	1.8	4.9	43.6	3.8	44.7	2.8	45.7	1.5	47.0	
			WB	3.6	26.6	1.6	0.8	2.5	27.2	1.6	27.7	0.7	28.2	-0.4	28.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.9	14.7	16.0	18.7	22.2	25.9	28.4	29.5	29.7	28.1	24.9	20.3	16.2	
	DBStd	5.91	2.13	2.79	3.39	3.56	2.99	2.23	1.62	1.41	1.78	2.33	2.50	2.19	
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	311	114	77	35	4	0	0	0	0	0	0	9	72	
	CDD10.0	4717	147	168	268	364	493	553	605	610	544	461	309	193	
	CDD18.3	1986	2	11	45	118	235	303	347	352	294	203	68	7	
	CDH23.3	22304	17	98	417	1209	2630	3727	4421	4468	3286	1664	338	31	
CDH26.7	11143	1	27	157	553	1348	2047	2396	2347	1590	606	68	4		
Wind	WSAvg	3.7	3.4	3.7	4.1	4.3	4.2	4.0	3.6	3.5	3.6	3.4	3.1	3.3	
Precipitation	PrecAvg	23	6	4	5	1	0	0	0	0	0	1	3	6	
	PrecMax	45	22	16	30	10	6	1	0	0	0	14	21	25	
	PrecMin	5	0	0	0	0	0	0	0	0	0	0	0	0	
	PrecStd	11	6	4	6	2	1	0	0	0	0	3	5	7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.4	30.2	35.0	39.1	40.9	40.9	40.0	38.2	39.0	36.0	31.7	26.7	
		MCWB	13.5	16.0	16.4	19.0	19.2	20.9	22.6	22.9	21.3	20.1	18.0	15.7	
	2%	DB	22.3	26.2	30.8	35.0	37.2	38.1	37.2	36.9	36.2	33.1	28.2	23.4	
		MCWB	12.9	14.2	15.9	17.9	19.3	21.3	22.7	22.9	21.8	20.4	18.0	15.1	
	5%	DB	20.4	23.3	27.8	32.2	35.1	36.2	36.0	35.8	34.4	31.5	26.7	21.9	
		MCWB	12.7	13.5	15.0	17.0	18.9	21.1	22.6	23.0	22.0	20.5	17.6	14.7	
	10%	DB	19.1	21.1	25.1	29.8	33.0	34.9	34.9	34.8	33.1	29.9	25.1	20.6	
		MCWB	12.4	12.8	14.6	16.5	18.7	21.2	22.6	22.9	21.9	20.3	17.1	14.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.9	17.4	19.5	20.8	22.0	24.7	26.1	26.5	25.2	23.9	21.4	18.2
			MCDB	21.1	26.0	28.2	32.3	33.4	32.9	32.8	32.3	30.4	28.8	25.9	22.2
2%		WB	14.7	15.7	17.4	19.3	21.0	23.6	25.1	25.6	24.3	22.6	20.3	16.8	
		MCDB	19.3	22.6	26.5	30.4	31.0	31.5	31.9	31.6	30.2	28.1	24.7	21.6	
5%		WB	13.8	14.6	16.3	18.2	20.3	22.9	24.5	25.0	23.6	21.9	19.2	15.8	
		MCDB	18.7	20.7	24.2	29.2	30.2	31.5	31.3	31.1	29.7	27.2	23.9	20.3	
10%		WB	13.1	13.7	15.3	17.2	19.6	22.3	24.0	24.5	23.1	21.3	18.3	14.8	
		MCDB	17.8	19.6	22.7	27.0	29.3	30.6	30.8	30.5	29.2	26.9	23.3	19.2	
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.2	10.3	11.8	12.5	12.0	10.9	10.3	10.3	9.6	9.1	8.5	
		MCDBR	10.3	12.5	14.0	15.5	14.9	13.8	12.1	11.4	11.8	11.2	10.6	9.9	
		MCWBR	5.2	5.3	5.3	5.5	5.0	4.1	3.9	3.7	3.6	3.6	4.4	5.0	
	5% WB	MCDBR	9.4	10.7	12.2	14.0	13.1	12.4	11.3	10.7	10.9	10.1	9.5	9.2	
		MCWBR	5.3	5.0	5.3	5.4	4.4	3.9	3.8	3.5	3.4	3.5	4.3	5.2	
Clear-Sky Solar Irradiance	taub	0.442	0.490	0.537	0.569	0.575	0.487	0.474	0.476	0.483	0.494	0.432	0.418		
	taud	1.966	1.845	1.736	1.693	1.710	1.946	1.994	1.990	1.962	1.934	2.069	2.067		
	Ebn at Noon	757	758	753	748	746	811	820	814	791	745	765	762		
	Edh at Noon	153	188	224	242	240	188	179	178	176	170	137	132		
All-Sky Solar Radiation	RadAvg	3.58	4.42	5.76	6.83	7.60	8.22	8.02	7.44	6.45	5.09	3.93	3.34		
	RadStd	0.11	0.20	0.18	0.22	0.24	0.09	0.06	0.08	0.11	0.12	0.13	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.60	N/A	-1.36	N/A	N/A	N/A	N/A	-50	+212	+171
(47)	Regional (6 neighbors)	+0.55	N/A	-0.65	+0.53	+0.37	N/A	N/A	-58	+199	+141

Nomenclature: See separate page